URBAN HEAT ISLAND ANALYSIS

Melbourne, Australia

Land Surface Temperature Mapping using Landsat 8 Thermal Remote Sensing

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Scene Date	23 February 2020 (Late Summer)
Satellite	Landsat 8 OLI/TIRS — Band 4, 5, 10
Study Area	Greater Melbourne GCCSA, Victoria, Australia
Tools	ArcGIS Pro (Spatial Analyst), USGS EarthExplorer
LST Range	13.24°C – 37.80°C

Executive Summary

This report presents a Land Surface Temperature (LST) analysis of Greater Melbourne derived from Landsat 8 Band 10 thermal infrared imagery acquired on 23 February 2020, representing Melbourne's late summer thermal peak. A six-step processing chain — from Top of Atmosphere Radiance through to emissivity-corrected LST — was implemented in ArcGIS Pro Spatial Analyst, producing spatially continuous surface temperature outputs at 30m resolution across the Greater Capital City Statistical Area.

Surface temperatures across Greater Melbourne ranged from 13.24°C to 37.80°C. The analysis identifies a pronounced Urban Heat Island (UHI) effect concentrated in the city's western and northern growth corridors, where high impervious surface density and limited canopy cover drive surface temperatures significantly above regional averages. The Yarra Valley fringe, Dandenong Ranges, and Port Phillip Bay coastline function as primary thermal buffers, registering the lowest LST values in the scene. NDVI analysis confirms a strong inverse relationship between vegetation cover and surface temperature, directly supporting the case for targeted urban greening in Melbourne's hottest suburbs.

These findings are directly relevant to the City of Melbourne Urban Forest Strategy and the Victorian Urban Cooling Strategy, both of which identify surface temperature reduction as a key climate adaptation priority.

1. Study Area & Context

Greater Melbourne is Australia's second-largest city, home to approximately 5.2 million people across a metropolitan area of roughly 9,990 km². The city's temperate oceanic climate (Köppen Cfb) produces warm to hot summers with high fire weather risk, and Melbourne regularly records extreme heat events during January and February. The February 2020 scene used in this analysis captures late-summer thermal conditions representative of Melbourne's annual peak heat loading.

Melbourne faces a well-documented and politically prominent urban heat challenge. The urban heat island effect — driven by the replacement of vegetation with roads, rooftops, and paved surfaces — amplifies surface temperatures in the city's western and northern growth corridors relative to the cooler, leafier inner-eastern suburbs. This spatial inequality in heat exposure is a central concern of both the City of Melbourne Urban Forest Strategy (target: 40% canopy cover by 2040) and the Victorian Government's Urban Cooling Strategy.

2. Methodology

All imagery was sourced from the USGS EarthExplorer platform (Landsat Collection 2 Level-1, LC08_L1TP_092086_20200223). Bands 4, 5, and 10 were downloaded and clipped to the Greater Melbourne GCCSA boundary (ABS, 2021). Processing was conducted in ArcGIS Pro Spatial Analyst Raster Calculator. The following six-step chain was applied:

Step	Output	Formula / Expression
1	TOA Radiance	L = 0.0003342 × B10 + 0.1
2	Brightness Temperature	BT = 1321.0789 / Ln(774.8853 / L + 1) − 273.15
3	NDVI	NDVI = (B5 − B4) / (B5 + B4)
4	Proportion of Vegetation	Pv = Square((NDVI − NDVImin) / (NDVImax − NDVImin))
5	Land Surface Emissivity	LSE = 0.004 × Pv + 0.986
6	Land Surface Temperature	LST = BT / (1 + (0.00115 × BT / 1.4388) × Ln(LSE))

Thermal constants applied: K1 = 774.8853 W/(m^2·sr·μm), K2 = 1321.0789 K (Landsat 8 Band 10 TIRS1). Band 11 was excluded in accordance with USGS guidance on Band 11 calibration uncertainty. All outputs were produced at 30m spatial resolution.

3. Results & Analysis

3.1 Land Surface Temperature

LST across Greater Melbourne ranged from 13.24°C to 37.80°C in the February 2020 late-summer scene. The spatial distribution reveals a clear west-east thermal gradient: Melbourne's western suburbs — including Sunshine, Werribee, and Melton — exhibit the highest surface temperatures, driven by high impervious surface fractions, sparse street tree canopy, and limited proximity to cooling water bodies. By contrast, the inner-eastern suburbs (Kew, Hawthorn, Brighton) and the Yarra Valley fringe register significantly lower LST values, reflecting higher vegetation density and the moderating influence of the Yarra River corridor.

The Melbourne CBD and inner-northern suburbs show elevated LST consistent with dense urban morphology, though the thermal signature is spatially heterogeneous due to the interspersed presence of parklands (Royal Park, Fitzroy Gardens, Albert Park Lake). Port Phillip Bay exerts a clear coastal cooling effect, with LST values along the foreshore substantially below the urban average — a pattern that has significant implications for planning the resilience of bayside communities under future heat scenarios.

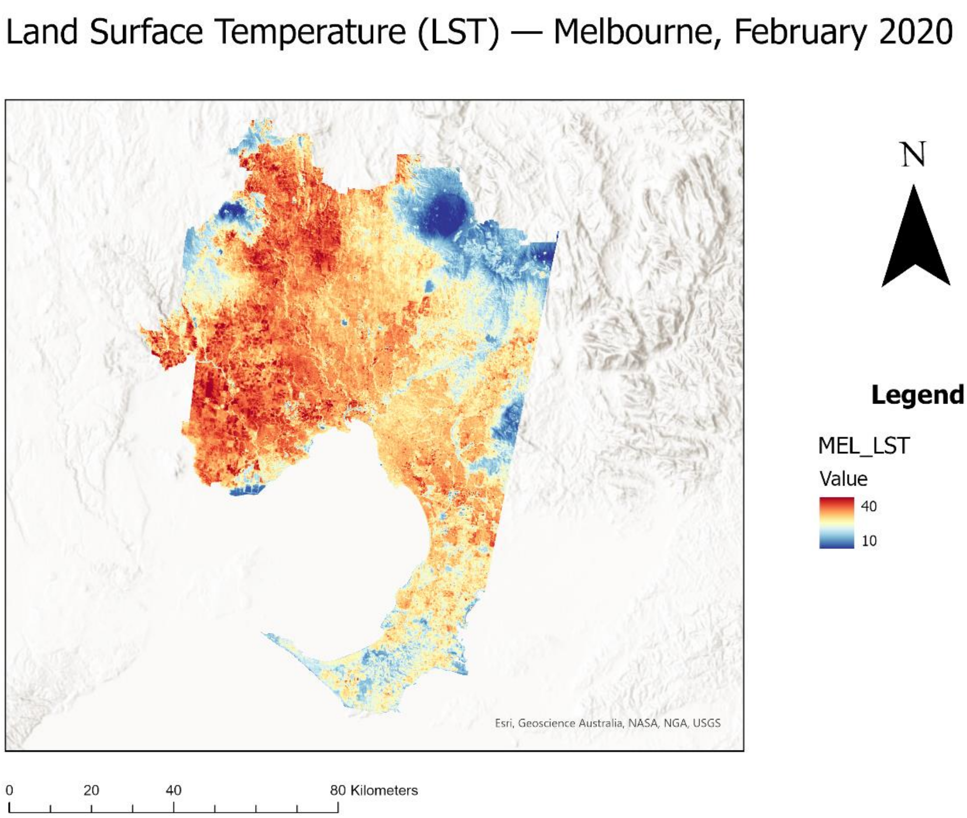

Figure 1: Land Surface Temperature — Greater Melbourne, February 2020 (13.24°C – 37.80°C)

3.2 NDVI & Vegetation Cover

The NDVI layer confirms a strong spatial inverse relationship between vegetation fraction and surface temperature across the study area. High NDVI values (approaching 0.6–0.7) are recorded in the Dandenong Ranges, Yarra Valley, and Plenty Gorge Park — areas that correspond directly to the lowest LST values in the scene. The western growth corridor (Melton, Wyndham) displays characteristically low NDVI values, reflecting the dominance of new residential development with minimal established vegetation cover and high impervious surface exposure.

This NDVI-LST relationship propagates through the processing chain: higher NDVI yields higher Proportion of Vegetation (Pv), which elevates Land Surface Emissivity (LSE), which in turn reduces the emissivity-corrected LST output. The result is a measurable, spatially consistent cooling effect in vegetated zones that directly quantifies the thermal benefit of Melbourne's existing green infrastructure — and the cost of its absence in heat-exposed growth areas.

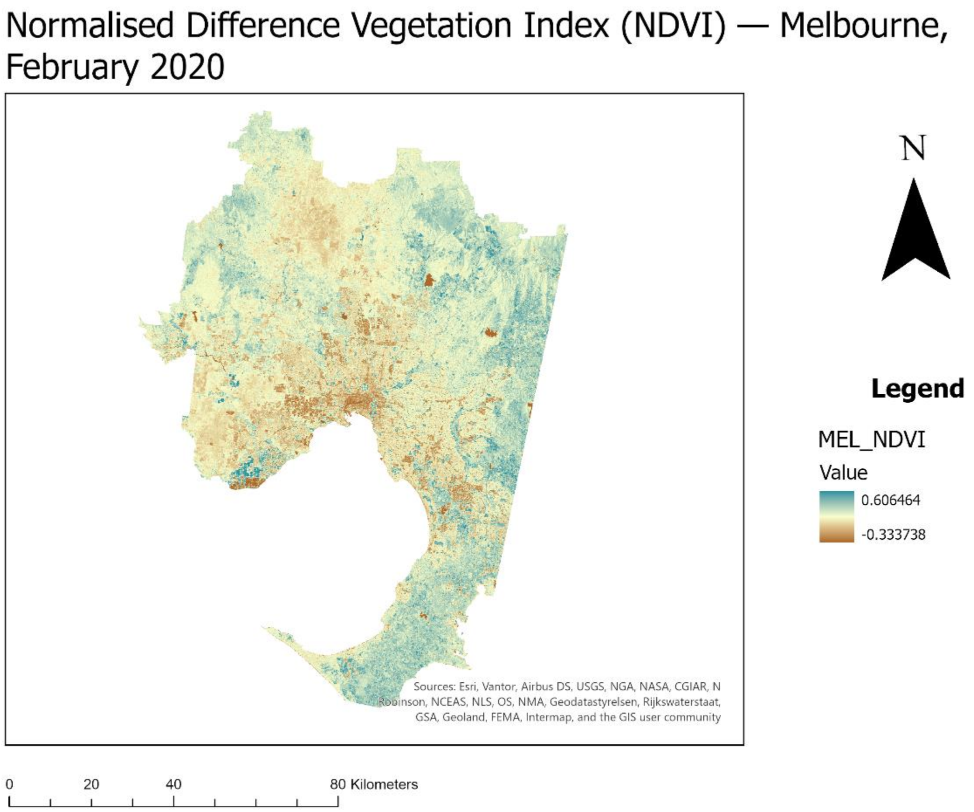

Figure 2: Normalised Difference Vegetation Index (NDVI) — Greater Melbourne, February 2020

4. Key Findings

- LST ranged from 13.24°C to 37.80°C across Greater Melbourne in February 2020, a thermal range of 24.56°C within a single metropolitan area.
- A pronounced west-east heat gradient is evident: Melbourne's western growth corridors (Sunshine, Werribee, Melton) consistently record the highest surface temperatures due to high impervious surface density and limited canopy cover.
- The Dandenong Ranges, Yarra Valley, and Port Phillip Bay coastline function as primary thermal buffers, registering the lowest LST values in the scene.
- NDVI is a strong inverse predictor of LST across the study area — suburbs with higher vegetation fraction demonstrate measurably lower surface temperatures, quantifying the cooling benefit of urban green infrastructure.
- The Yarra River corridor produces a visible linear cooling effect through the inner suburbs, consistent with the latent heat buffering effect of open water and riparian vegetation.
- The Melbourne CBD exhibits elevated but spatially heterogeneous LST, reflecting the thermal contrast between dense impervious surfaces and embedded parklands.

5. Policy Implications

The spatial distribution of surface temperatures identified in this analysis has direct relevance to Melbourne's existing urban heat policy frameworks:

City of Melbourne Urban Forest Strategy (40% canopy cover target by 2040): This analysis provides spatial evidence of which areas would benefit most from accelerated canopy investment. The western growth corridor — where surface temperatures are highest and NDVI is lowest — represents the priority zone for street tree planting and green infrastructure development.

Victorian Urban Cooling Strategy: The pronounced west-east thermal gradient identified here supports the strategy's focus on heat equity — western Melbourne communities face disproportionately high surface heat exposure relative to inner-eastern suburbs with higher canopy cover and greater proximity to cooling water bodies.

Urban Growth Boundary Planning: The Melton and Wyndham growth areas show the highest LST values and lowest NDVI in the scene. As these areas continue to develop, mandating minimum canopy cover ratios and cool surface materials in planning approvals would directly address the thermal outcomes quantified in this analysis.

6. Limitations

- Landsat 8 Band 10 has a native spatial resolution of 100m, resampled to 30m in the Level-1 product. This resampling introduces sub-pixel averaging that may smooth fine-scale thermal variation in heterogeneous urban environments.
- The single-date scene captures a snapshot of thermal conditions on one day in February 2020 and does not account for diurnal or inter-annual variability. A multi-temporal analysis would provide a more robust characterisation of Melbourne's heat exposure.
- The emissivity model applied ($LSE = 0.004 \times Pv + 0.986$) is a simplified two-endmember approach. More sophisticated emissivity retrieval methods (e.g. TES algorithm) may improve accuracy in spectrally complex urban environments.
- Atmospheric correction was not applied beyond the TOA Radiance conversion. Full atmospheric correction using MODTRAN or similar would improve absolute accuracy of derived LST values.

Data Sources

Landsat 8 Imagery	USGS EarthExplorer — LC08_L1TP_092086_20200223 (Collection 2 Level-1)
Study Area Boundary	ABS Greater Capital City Statistical Areas (GCCSA) — Greater Melbourne, 2021
Processing Software	Esri ArcGIS Pro 3.x — Spatial Analyst Extension, Raster Calculator
GitHub Repository	github.com/lukek6892/urban-heat-island-melbourne-landsat8